Macroeconomics #2

The Solow–Swan Growth Model

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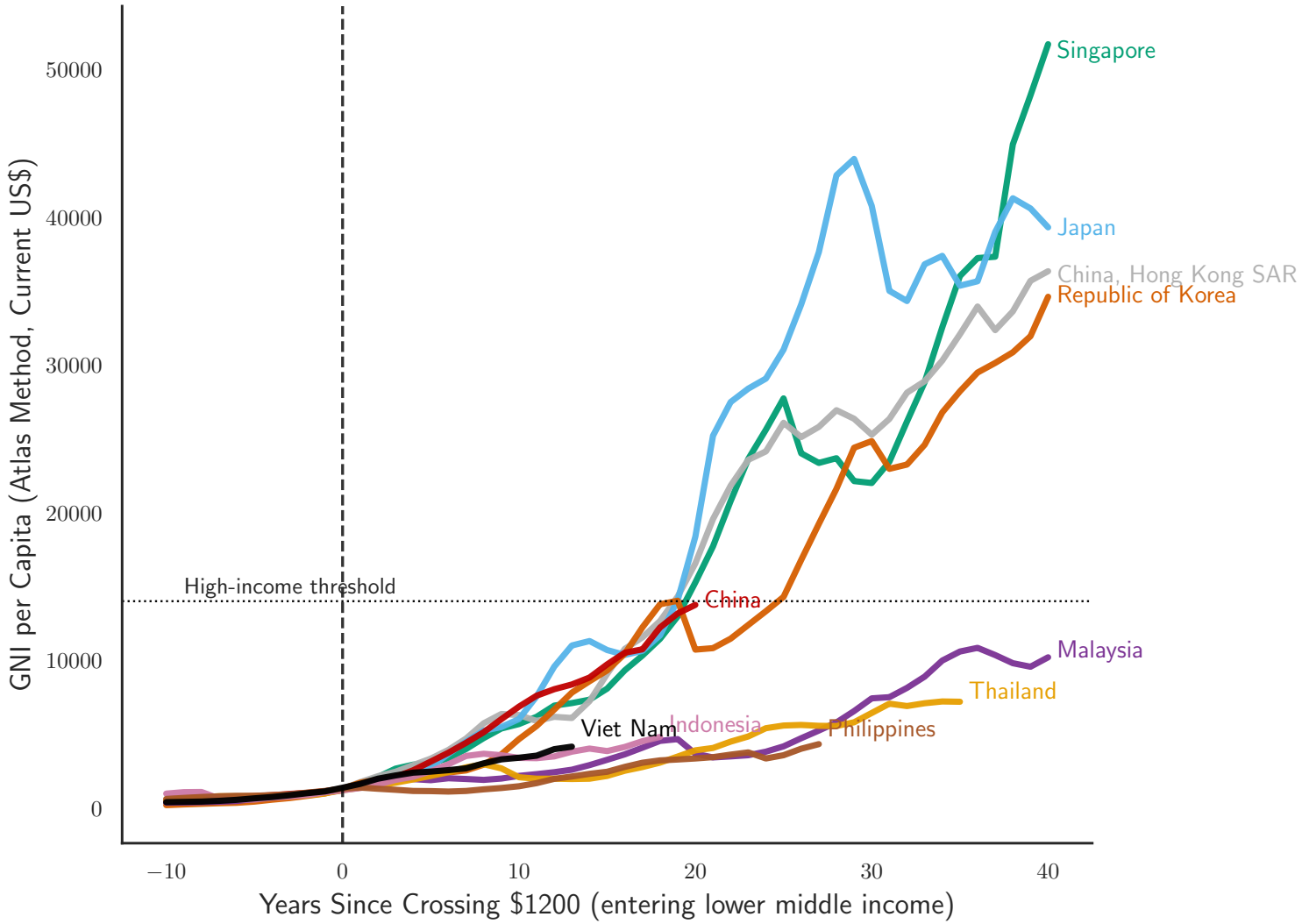
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Today, you will learn...

- the closed-economy Solow model
- how a model is used to explain cross-country income differences
- how a country's standard of living depends on its saving and population growth rates

Growth


GNI per Capita (Atlas Method, Current US$)
50000
40000
30000
20000
10000
0
High-income threshold
Singapore
Japan
China, Hong Kong SAR
Republic of Korea
China
Malaysia
Thailand
Viet Nam
Indonesia
Philippines
−10
0
10
20
30
40
Years Since Crossing $1200 (entering lower middle income)

The Solow growth model

- Introduced by Robert Solow [1] in 1956 (Nobel Prize in 1987)
- ignores government spending (G), taxes (T) and the external sector (no import/export)
- focuses on capital K and labor L (no unemployment)
- let capital and labor accumulate over time

[1] Solow, R. M. (1956). A contribution to the theory of economic growth. The Quarterly Journal of Economics, 70(1), 65-94.

The production function

Aggregate

$$Y = F(K, L)$$

Define

$$y = Y/L \qquad \text{output per worker}$$
$$k = K/L \qquad \text{capital per worker}$$

Assume constant returns to scale

$$zY = F(zK, zL) \qquad (z > 0)$$

To convert the production to per-worker level, set $z = 1/L$

$$Y/L = F(K/L, 1)$$
$$y = F(k, 1)$$
$$y = f(k)$$

Example

Cobb-Douglas production function
Aggregate

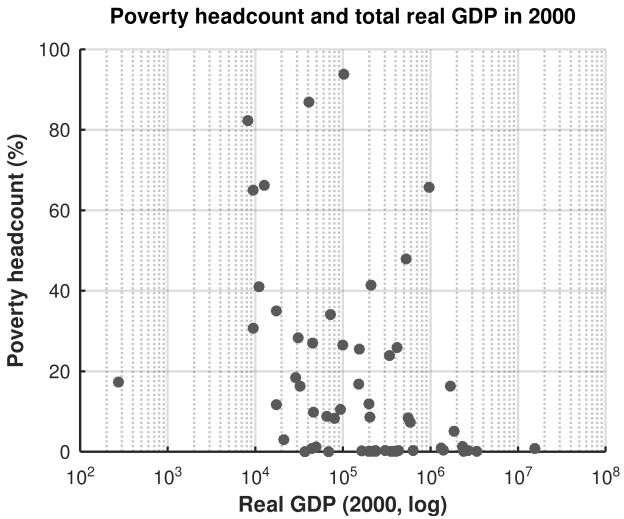
$$Y = F(K, L) = K^{\alpha} L^{1-\alpha}$$

Dividing both sides by L

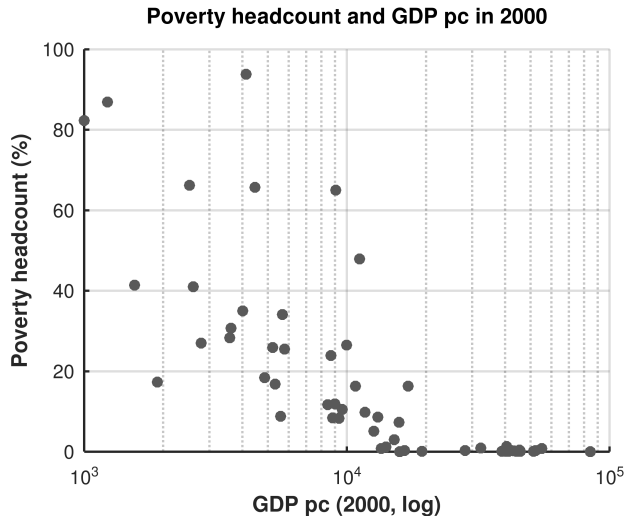
$$\frac{Y}{L} = \frac{K^{\alpha} L^{1-\alpha}}{L}$$
$$\frac{Y}{L} = K^{\alpha} L^{-\alpha}$$
$$\frac{Y}{L} = \left(\frac{K}{L}\right)^{\alpha}$$
$$y = k^{\alpha} = f(k)$$

Why do we care about per capita instead of aggregate?

- India has a big population and is the third-largest economy (in PPP terms)
- ...but poverty headcount ratio ($< \$8.3/day$) in 2021 is 82.1%!
- ... while US (2%), Japan (3.2%), China & Viet Nam (21.5%)



(a) Aggregate GDP



(b) GDP per capita

Source: wikipedia

National Income

Aggregate:

$$Y = C + I$$

Assume that every period, a fraction of national income s is saved

$$I = sY$$

These savings (investment) carry over to the next period as inputs to production.

Capital Accumulation

- Assume that population grows at the rate of n

$$L_{t+1} = (1+n)L_t$$

- Investment is saved from this period and is used in the next

$$K_{t+1} = I_t + K_t - \delta K_t$$

where δ is the depreciation rate: the fraction of the capital stock that wears out each period. Dividing both sides by L_t

$$\frac{K_{t+1}}{L_t} = \frac{sY_t}{L_t} + \frac{(1-\delta)K_t}{L_t}$$

$$\frac{K_{t+1}(1+n)}{L_{t+1}} = sy_t + (1-\delta)k_t$$

$$(1+n)k_{t+1} = sf(k_t) + (1-\delta)k_t$$

move $1+n$ to the other side

$$k_{t+1} = \frac{sf(k_t) + (1-\delta)k_t}{1+n} \tag{1}$$

Change in capital per worker

We are interested in the change in capital per worker over time. From (1), subtract k_t from both sides

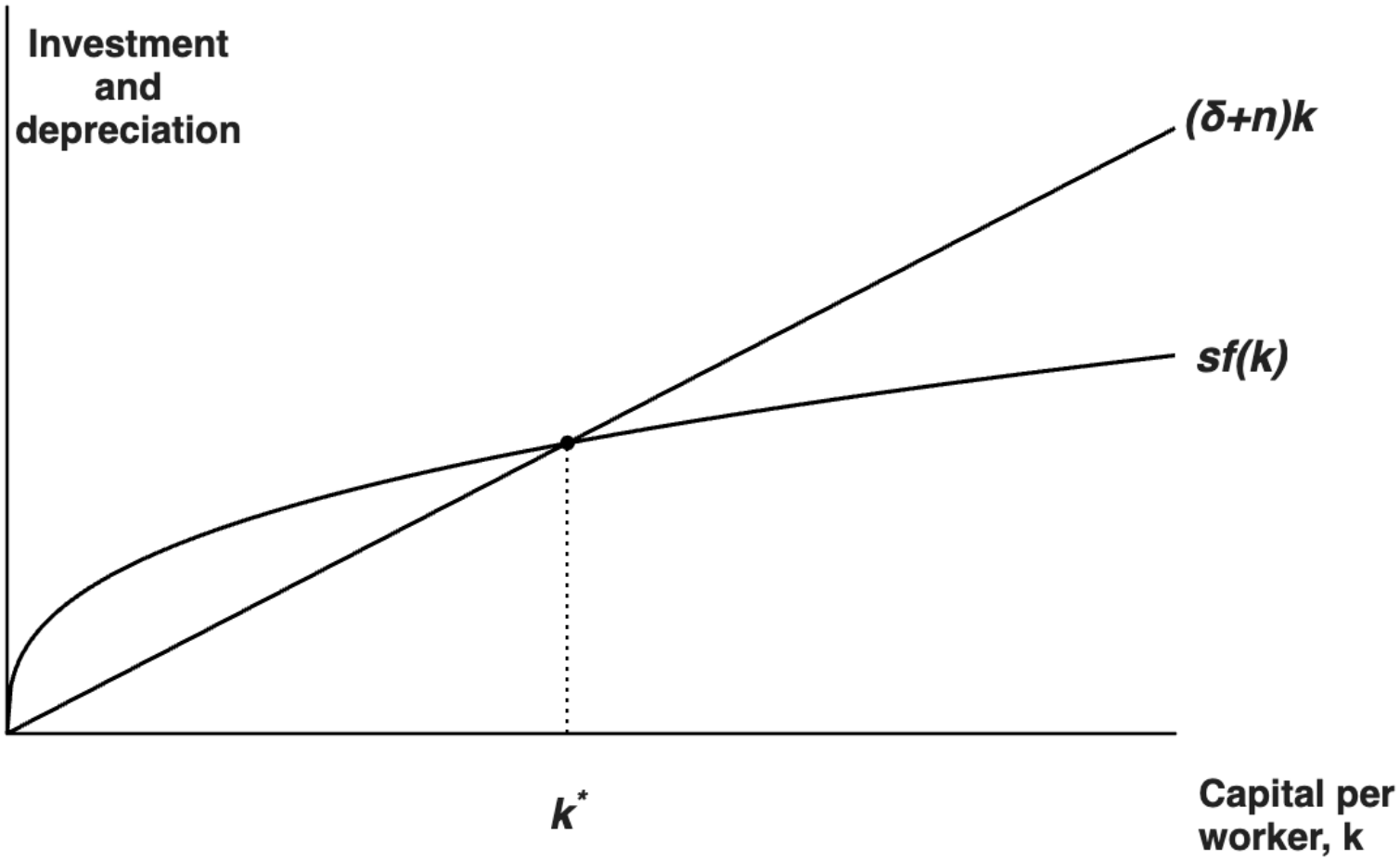
$$k_{t+1} - k_t = \frac{sf(k_t) + (1-\delta)k_t}{1+n} - k_t$$

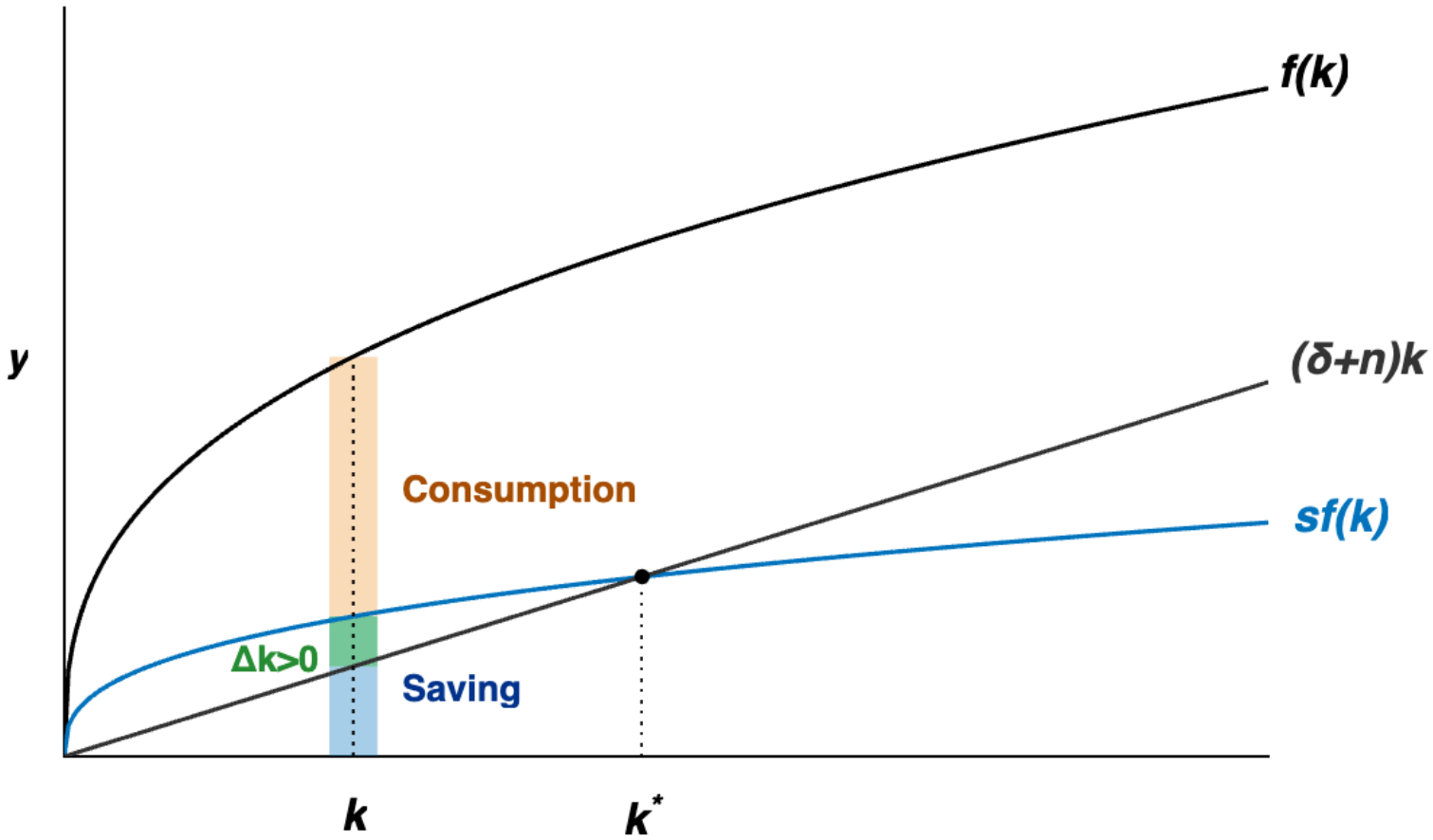
$$\Delta k_t = \frac{sf(k_t) + (1-\delta)k_t - (1+n)k_t}{1+n}$$

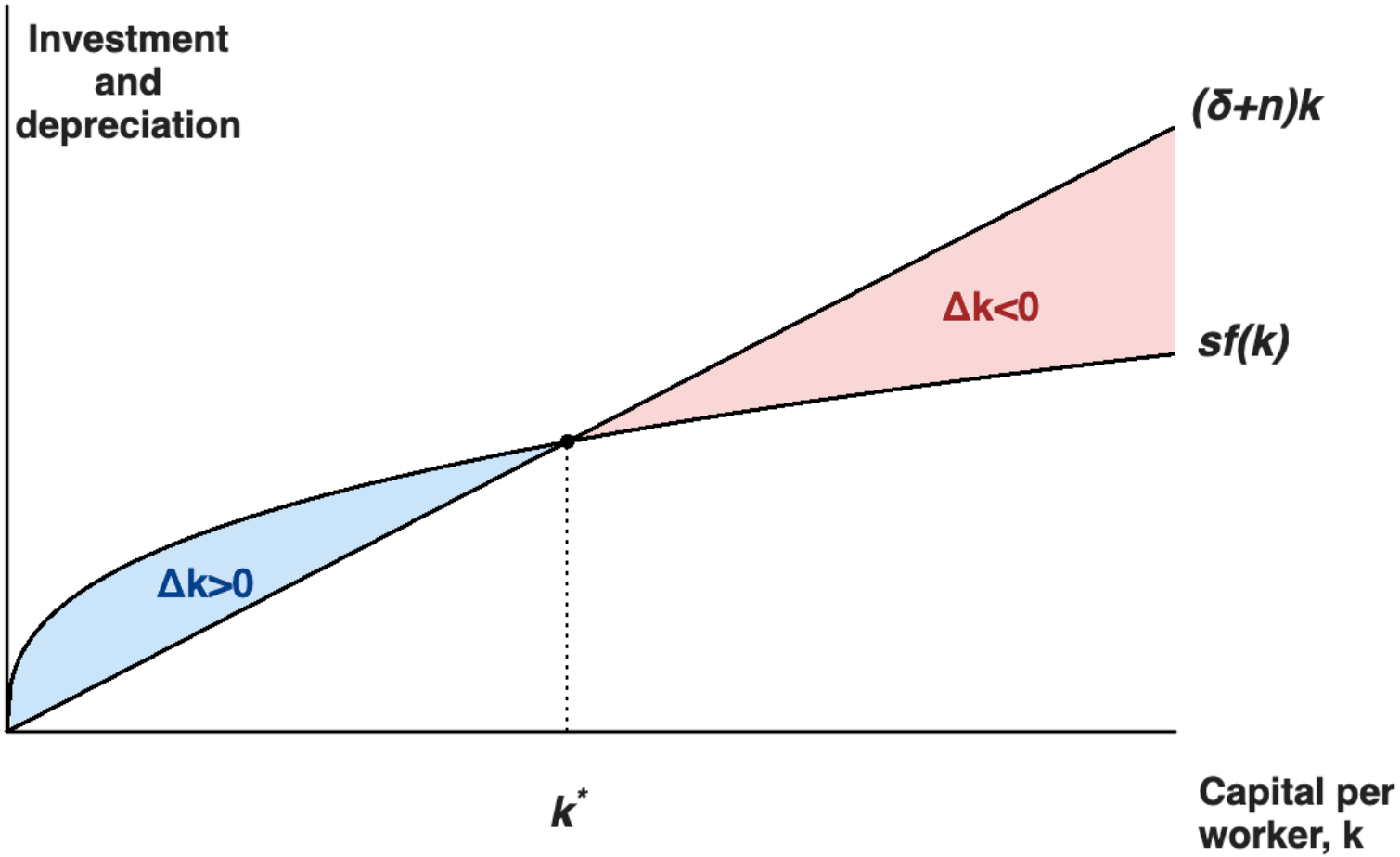
$$\Delta k_t = \frac{sf(k_t) - (\delta+n)k_t}{1+n} \tag{2}$$

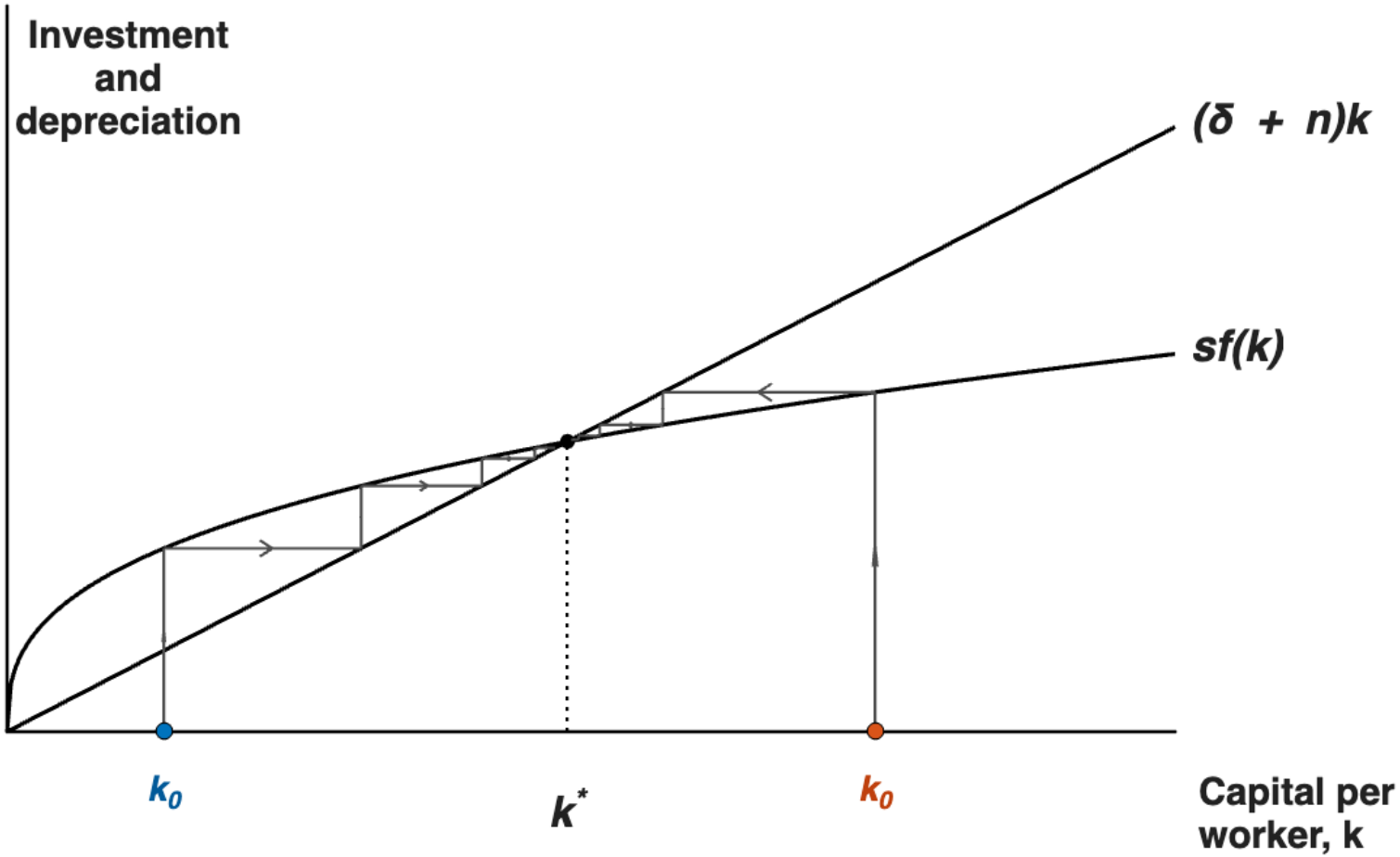
- high saving rate s increases capital accumulation
- high depreciation rate δ reduces it
- high population growth n reduces it

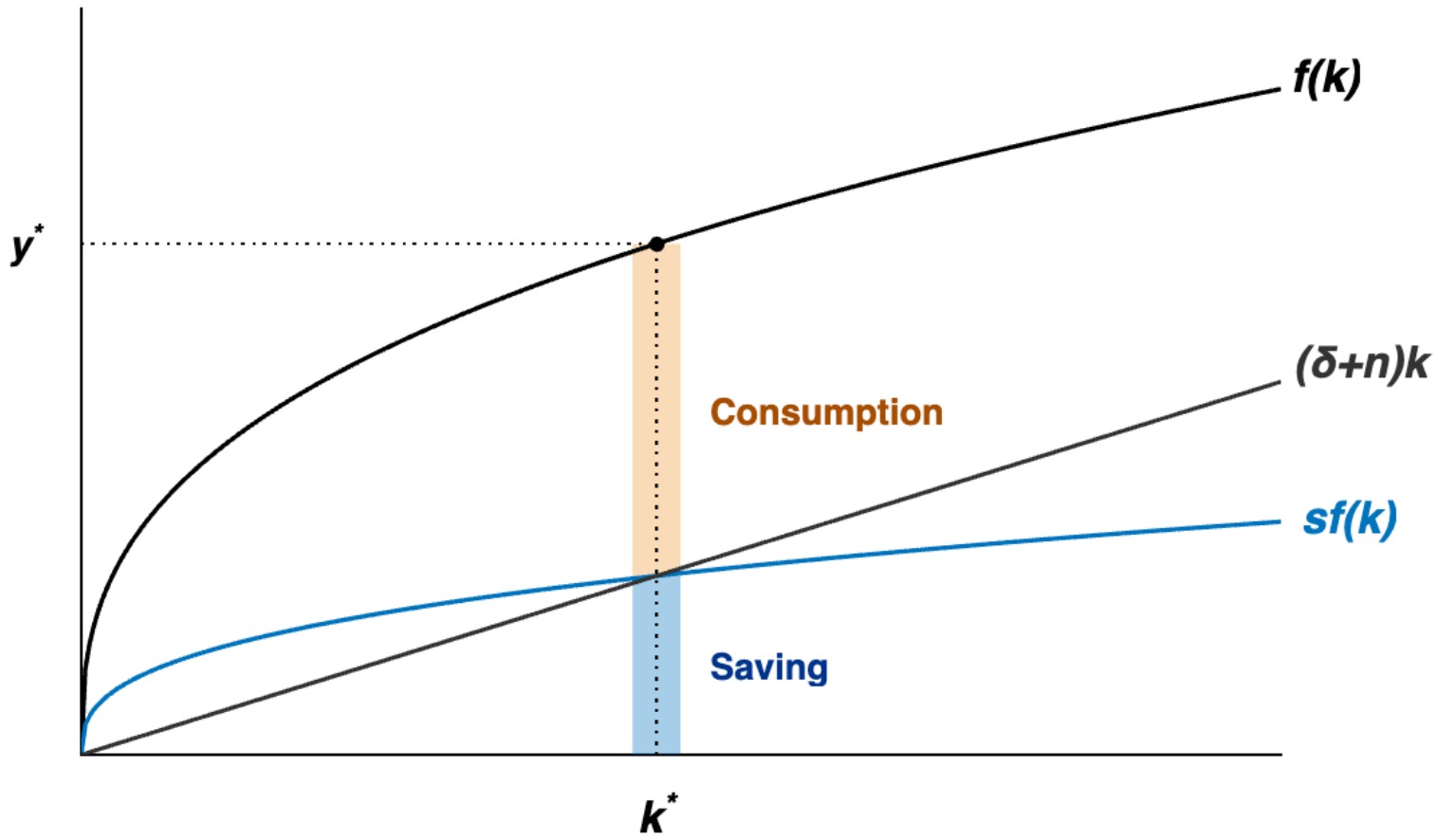
Graphical Representation




f(k)
y
(δ+n)k
Consumption
sf(k)
Δk>0
Saving
k
k*


Investment
and
depreciation
(δ+n)k
Δk<0
sf(k)
Δk>0
k*
Capital per
worker, k


Investment
and
depreciation
(δ + n)k
sf(k)
k0
k*
k0
Capital per
worker, k


f(k)
y*
(δ+n)k
Consumption
sf(k)
Saving
k*

Example

Let's assume the Cobb-Douglas production function

$$F(K, L) = K^{\alpha} L^{1-\alpha}$$

In per-worker term

$$y = f(k) = k^{\alpha}.$$

Then (1) becomes

$$k_{t+1} = \frac{sk_t^{\alpha} + (1-\delta)k_t}{1+n}$$

and (2) becomes

$$\Delta k_t = \frac{sk_t^{\alpha} - (\delta+n)k_t}{1+n}$$

Example

Assume: $\alpha = 0.36$, $n = 0.05$, $\delta = 0.10$ and $s = 0.20$

Low initial capital: $k_1 = 0.3$

t	k_t	k_{t+1}	Δk_t	growth (%)
1	0.3000	0.3806	+0.08063	26.88
2	0.3806	0.4608	+0.08016	21.06
3	0.4608	0.5391	+0.07829	16.99
5	0.6145	0.6866	+0.07206	11.73
10	0.9357	0.9880	+0.05230	5.59
20	1.3101	1.3329	+0.02277	1.74
40	1.5286	1.5321	+0.00355	0.23
60	1.5618	1.5623	+0.00060	0.03
∞	1.5675	1.5675	0	0

High initial capital: $k_1 = 2.0$

t	k_t	k_{t+1}	Δk_t	growth (%)
1	2.0000	1.9587	−0.04125	-2.06
2	1.9587	1.9216	−0.03719	- 1.90
3	1.9216	1.8880	−0.03354	- 1.75
5	1.8577	1.8304	−0.02734	-1.47
10	1.7449	1.7284	−0.01653	-0.95
20	1.6346	1.6285	−0.00618	- 0.38
40	1.5773	1.5764	−0.00090	- 0.06
60	1.5690	1.5688	−0.00001	- 0.01
∞	1.5675	1.5675	0	0

- Both paths approach the same **steady state**, $k^* \approx 1.5675$.
- At k^*, $\Delta k = 0$. Period 70 is still an approximation.
- How can we find this value without iterating?

How to locate the steady-state?

Set $\Delta k = 0$ in (2) and solve for k

$$\begin{aligned} sk^{\alpha} &= (\delta + n)k \\ s &= (\delta + n)k^{1-\alpha} \\ k^{1-\alpha} &= \frac{s}{\delta + n} \end{aligned}$$

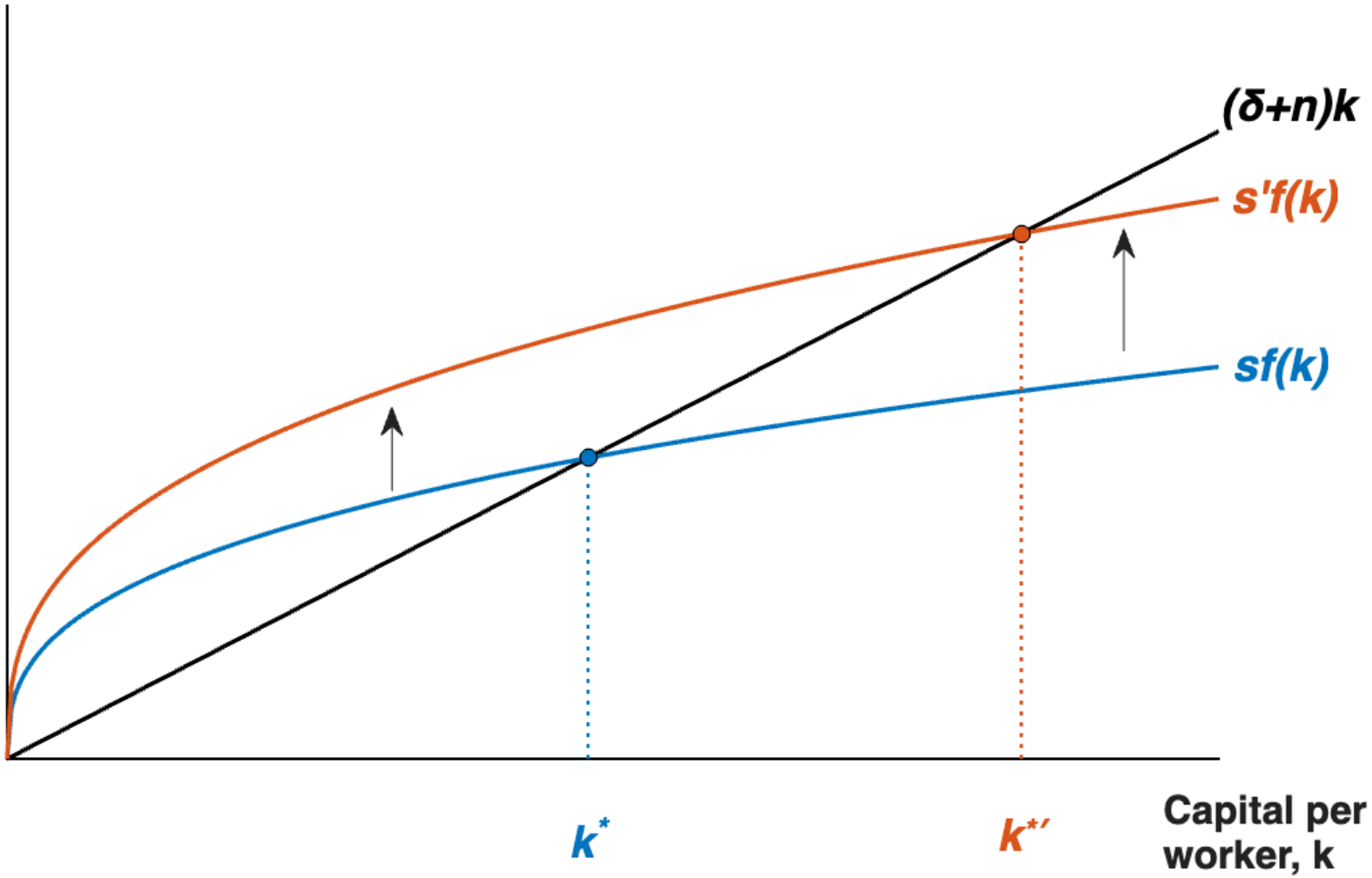
Exponentiate both sides by $1/(1-\alpha)$

$$k^* = \left(\frac{s}{\delta + n}\right)^{\frac{1}{1-\alpha}} \tag{3}$$

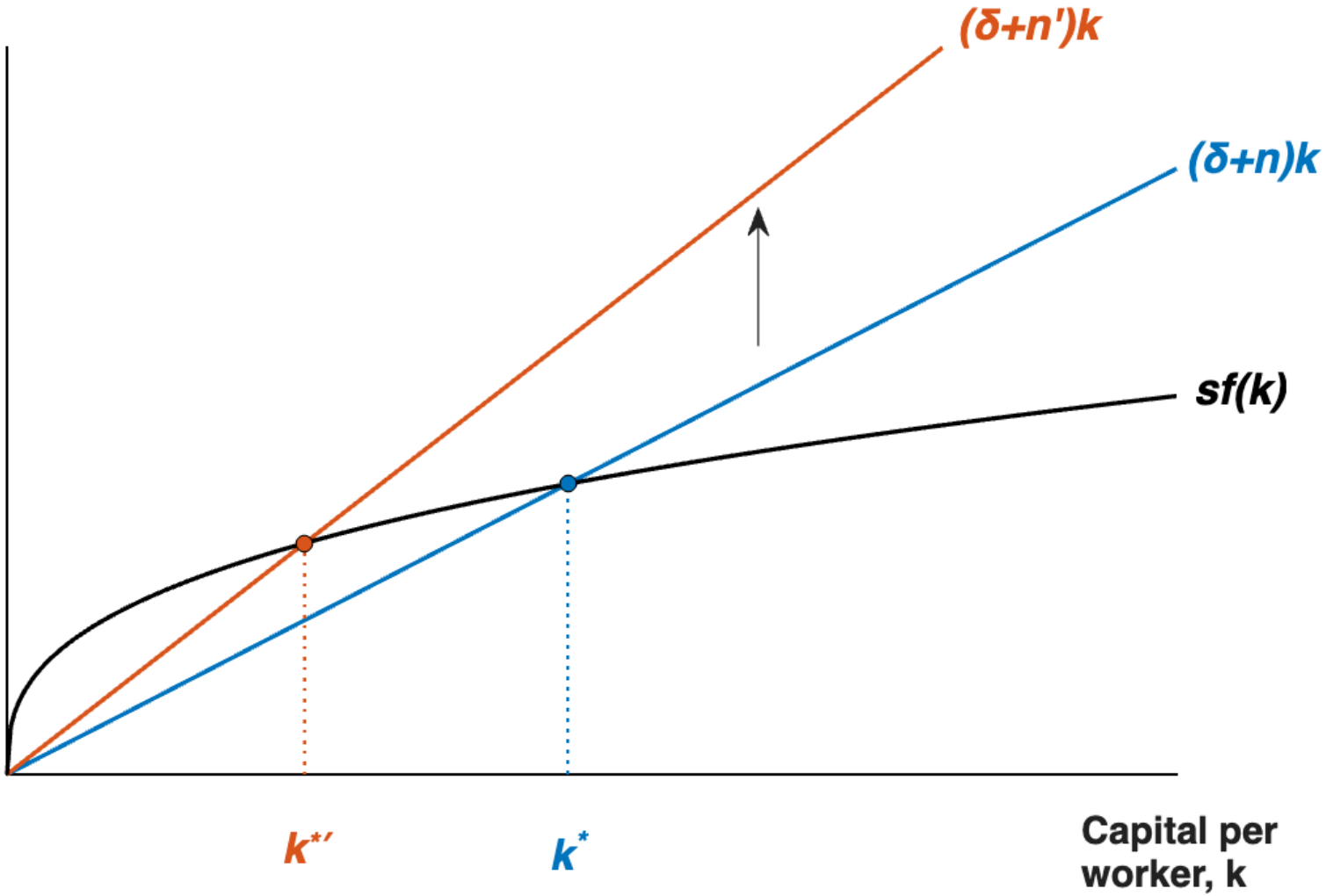
Solow's Predictions

1. A country is richer if it has high saving rates
2. A country is poorer if it has high population growth and a high depreciation rate.
3. Countries with the same s, n, δ and technology converge to the same steady state, whether they start at low or high income (*conditional* convergence)

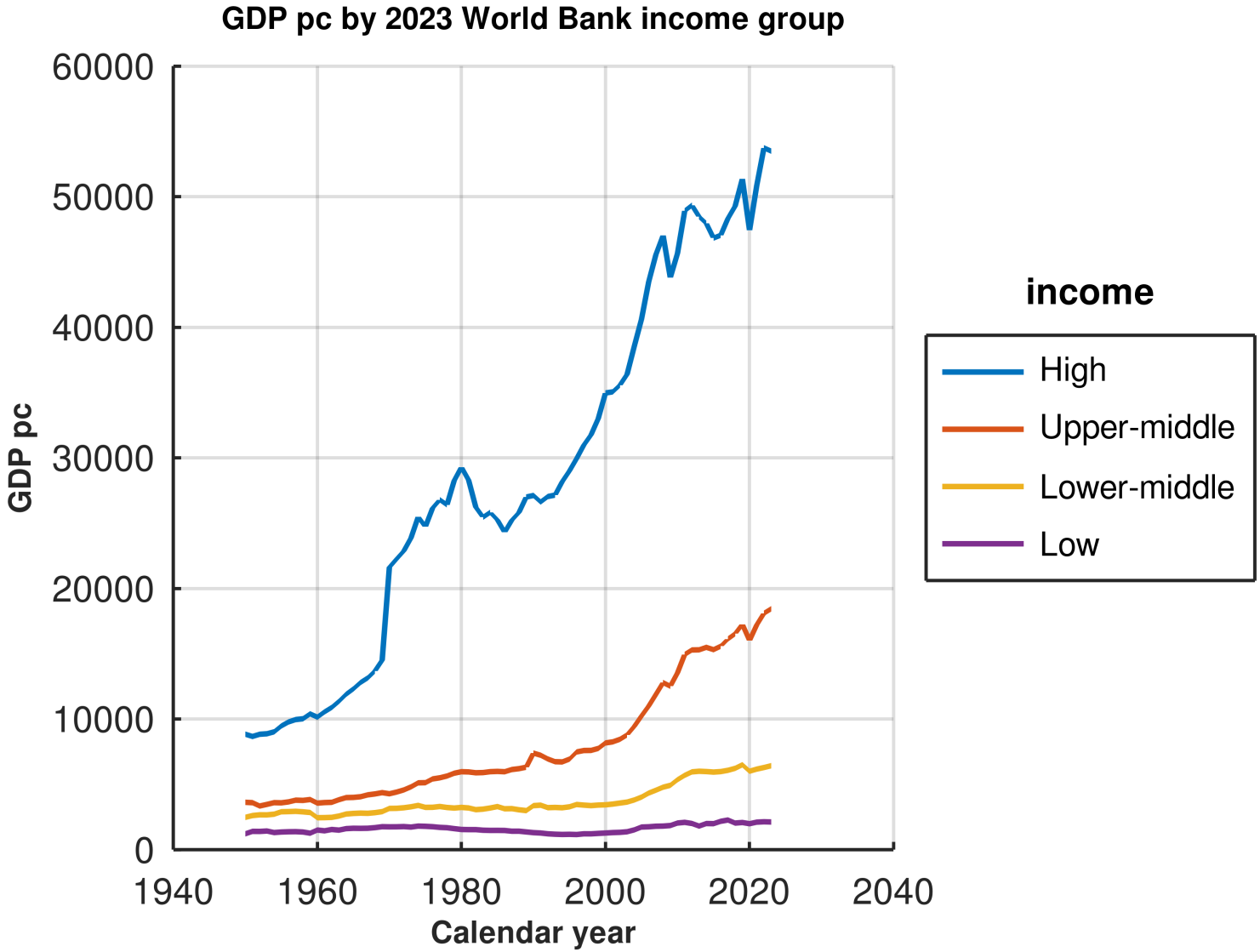
an increase in savings s

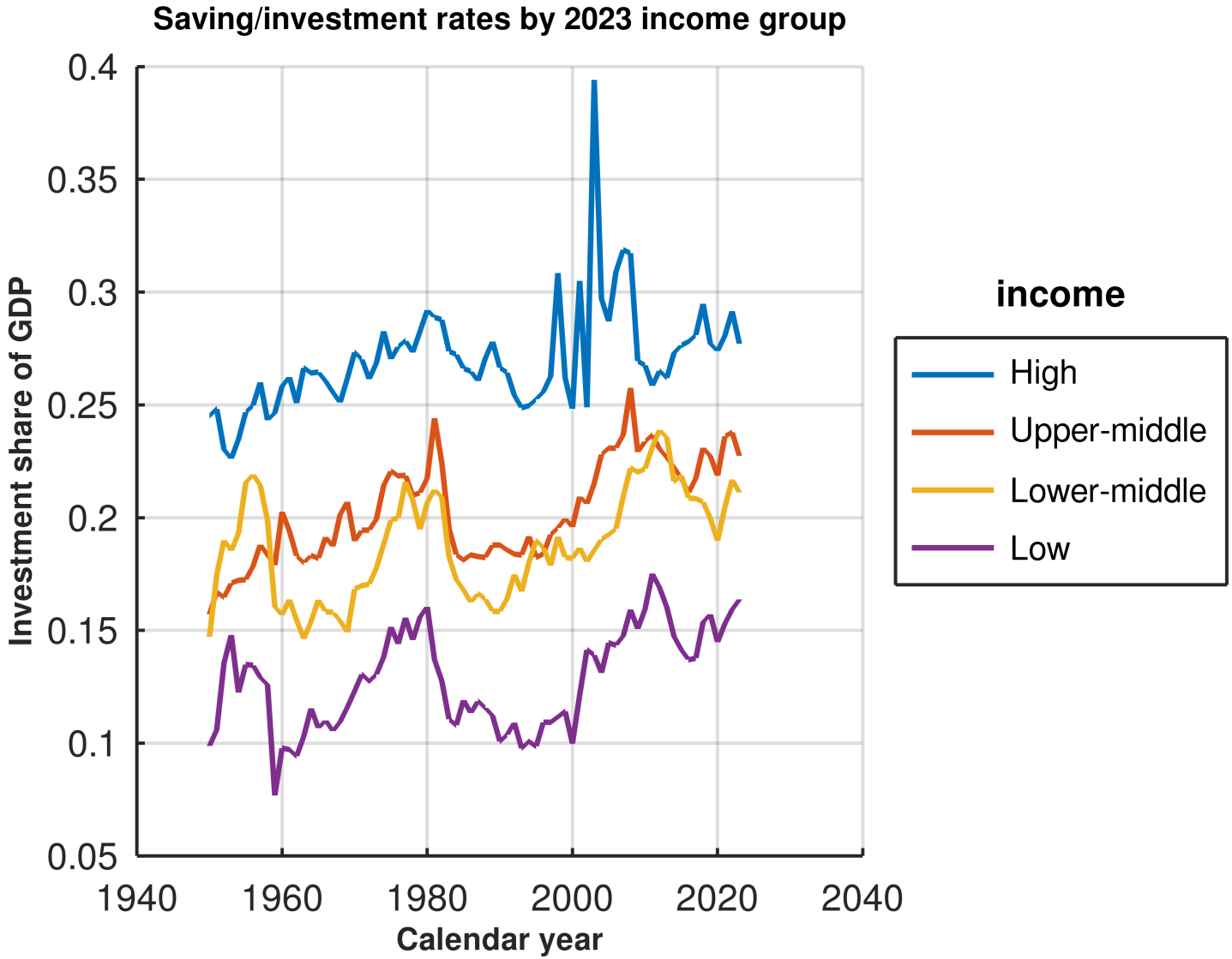


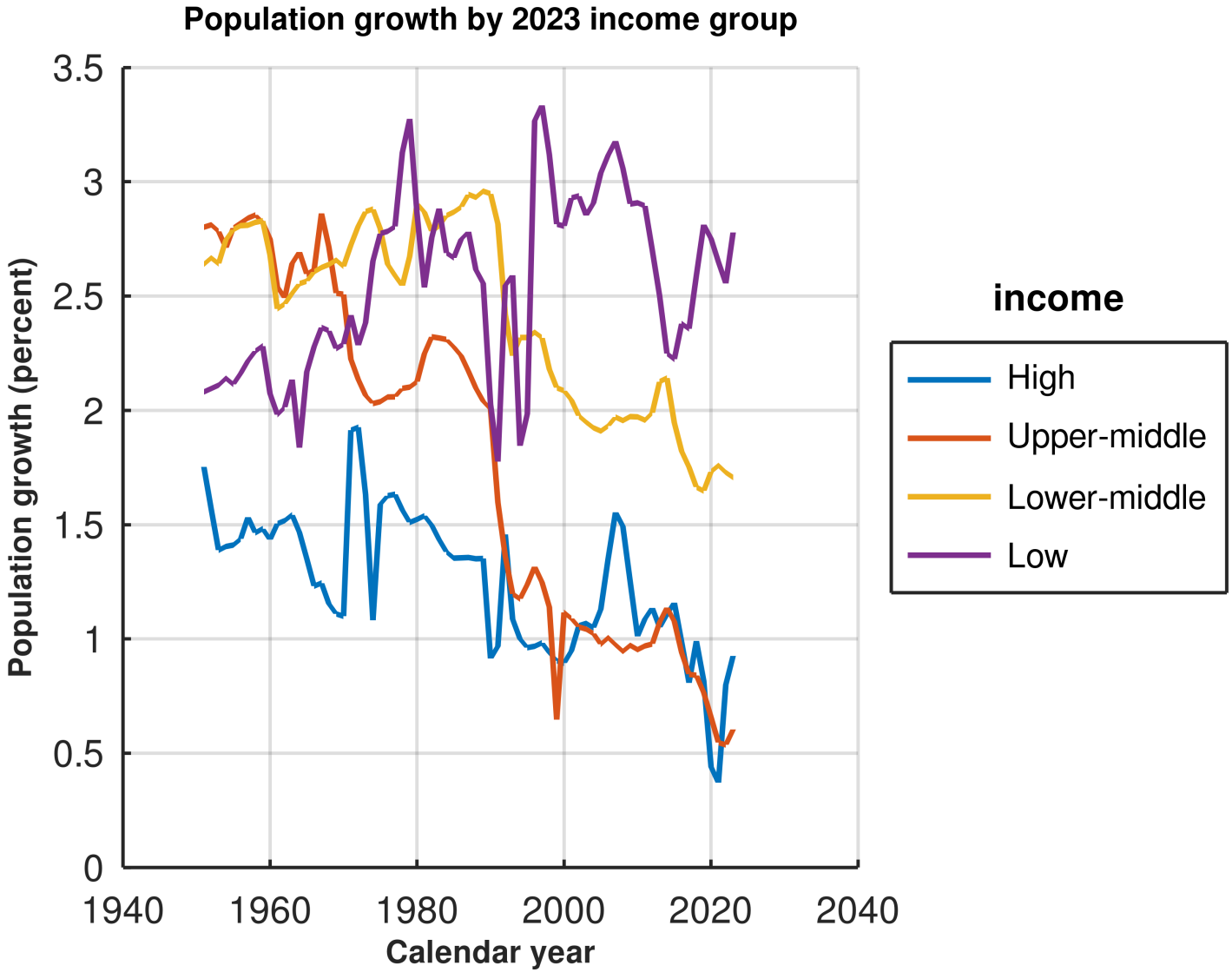
an increase in population growth n (or dep. rate δ)

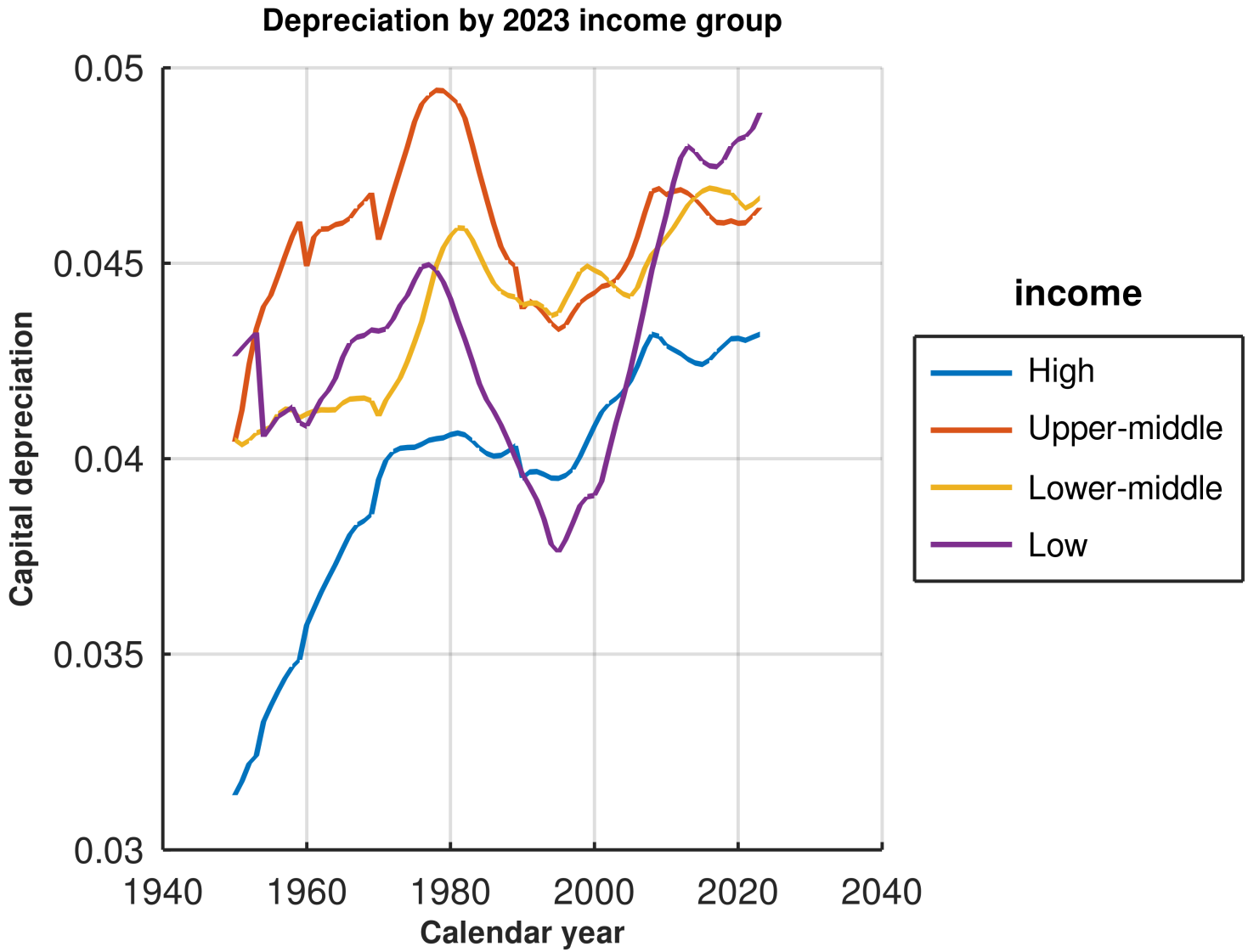


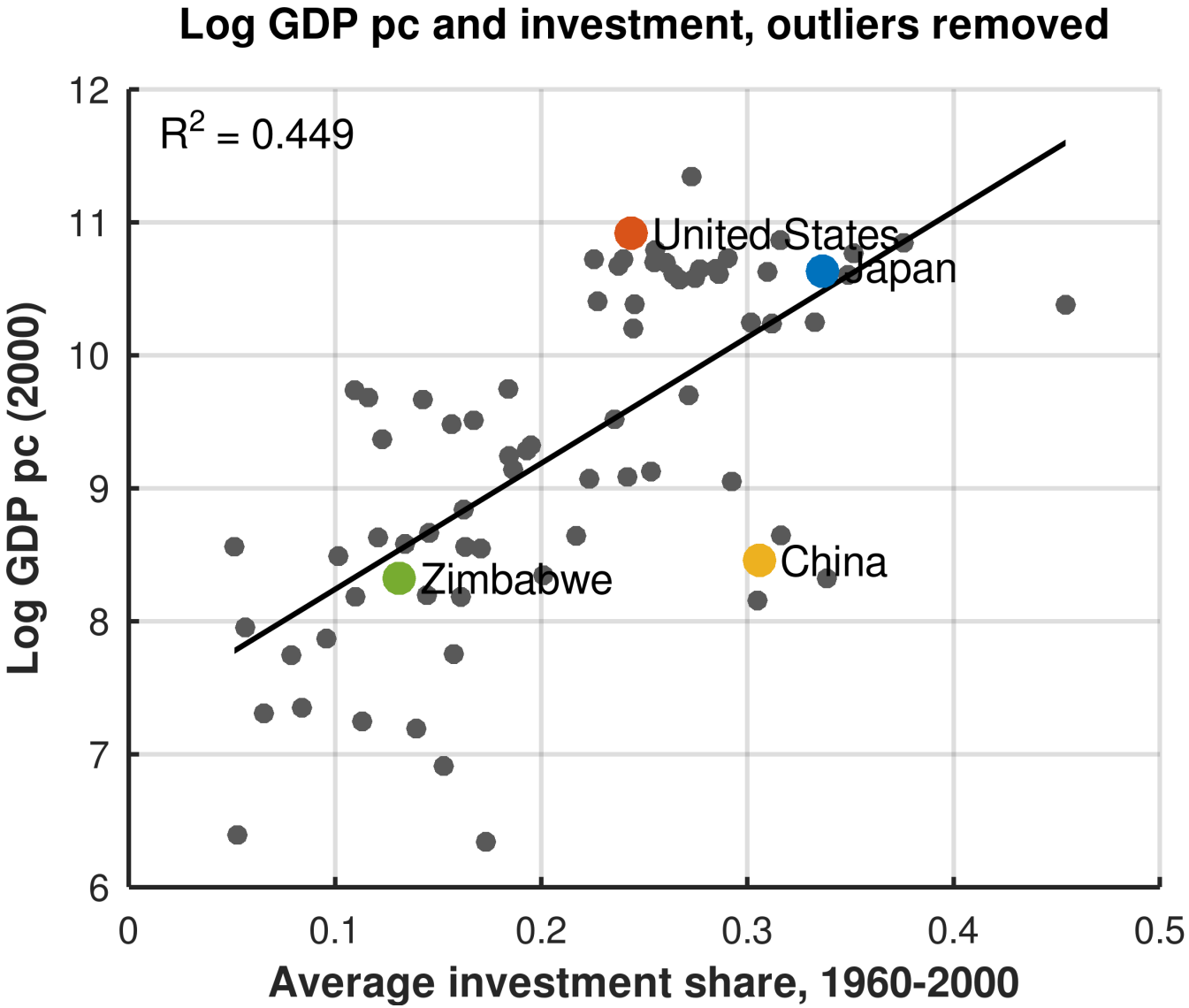
Evidence of Solow's prediction

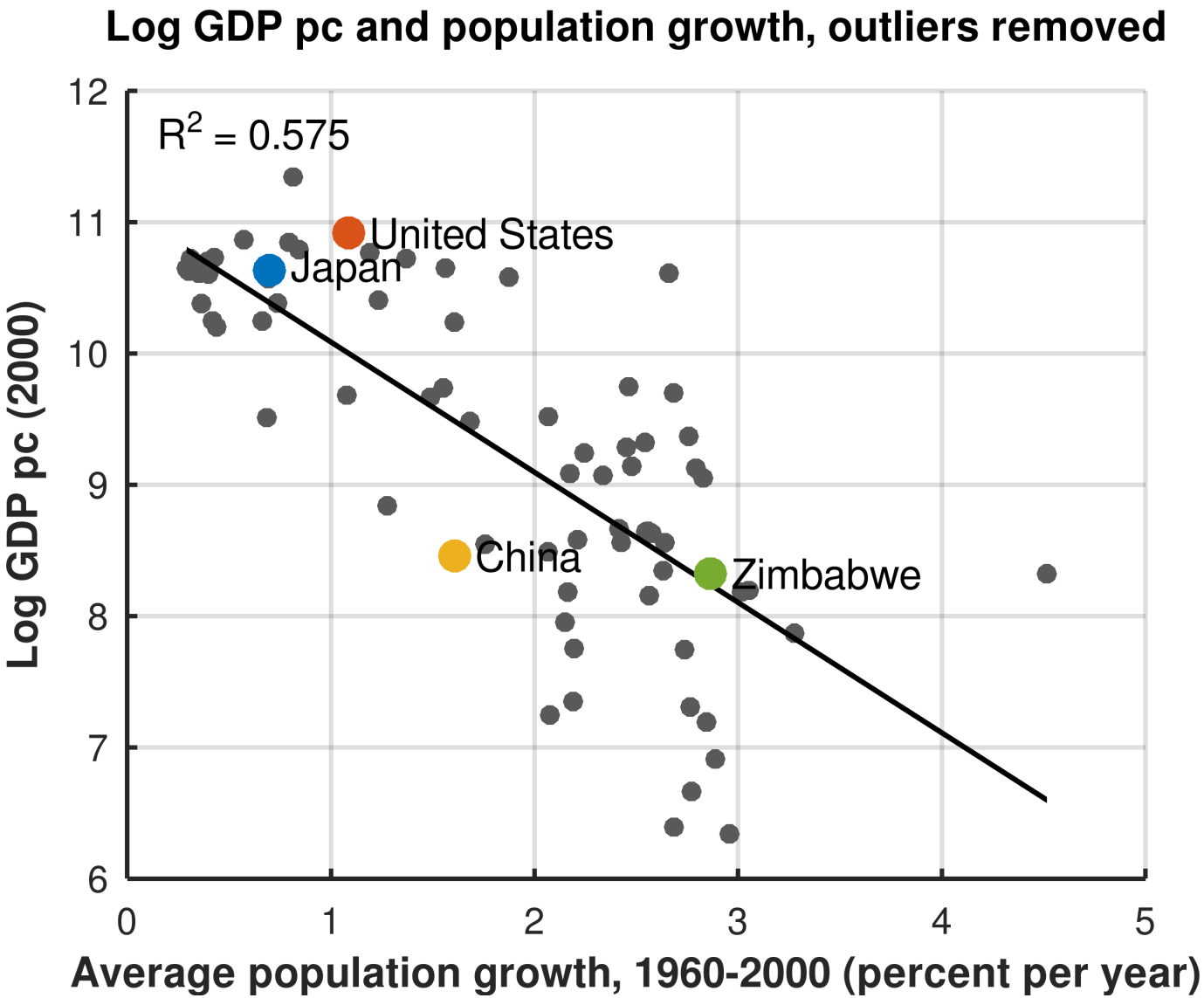

GDP pc by 2023 World Bank income group
GDP pc
60000
50000
40000
30000
20000
10000
0
1940
1960
1980
2000
2020
2040
Calendar year
income
High
Upper-middle
Lower-middle
Low

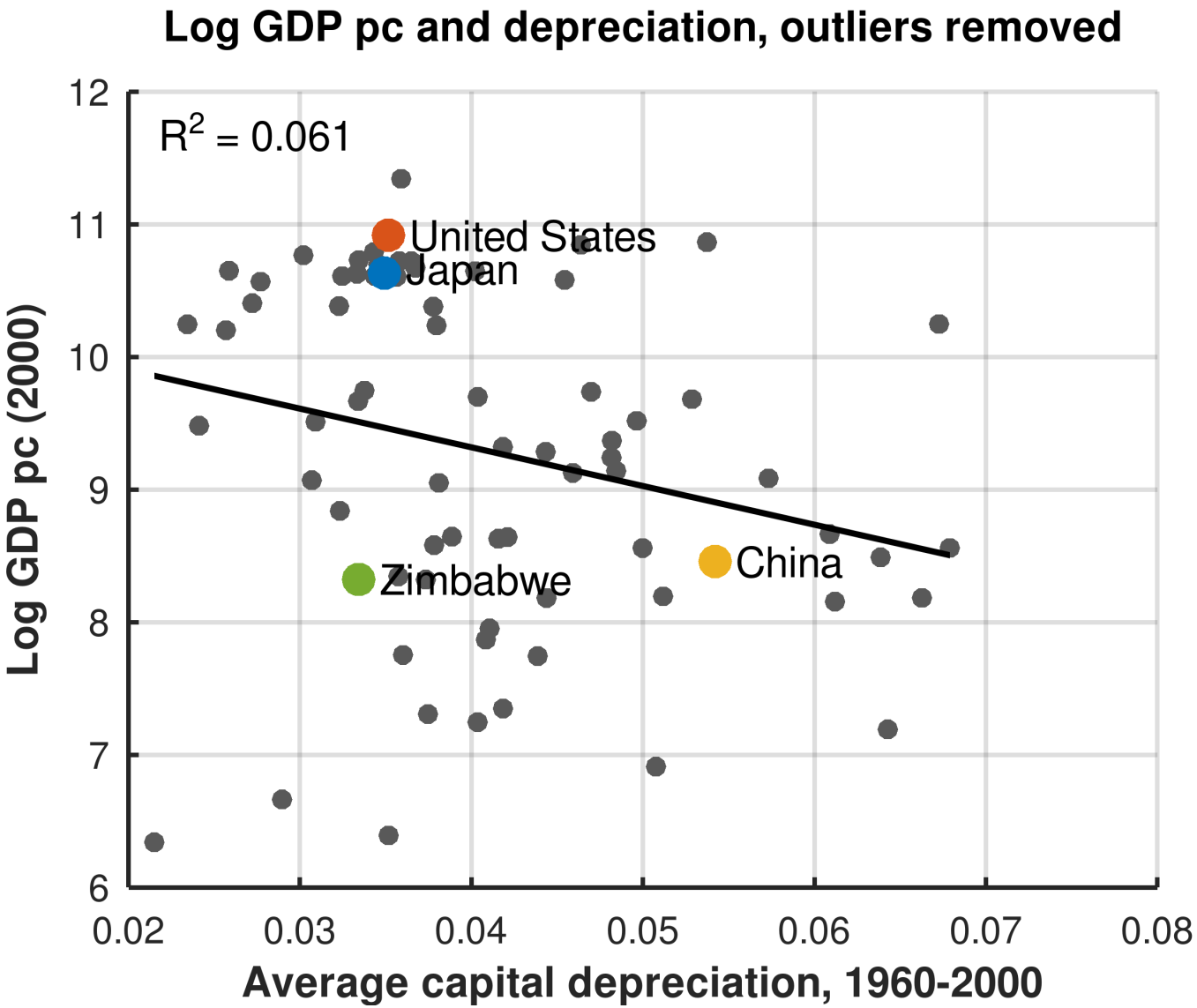

Saving/investment rates by 2023 income group
Investment share of GDP
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
1940
1960
1980
2000
2020
2040
Calendar year
income
High
Upper-middle
Lower-middle
Low


Population growth by 2023 income group
Population growth (percent)
3.5
3
2.5
2
1.5
1
0.5
0
1940
1960
1980
2000
2020
2040
Calendar year
income
High
Upper-middle
Lower-middle
Low

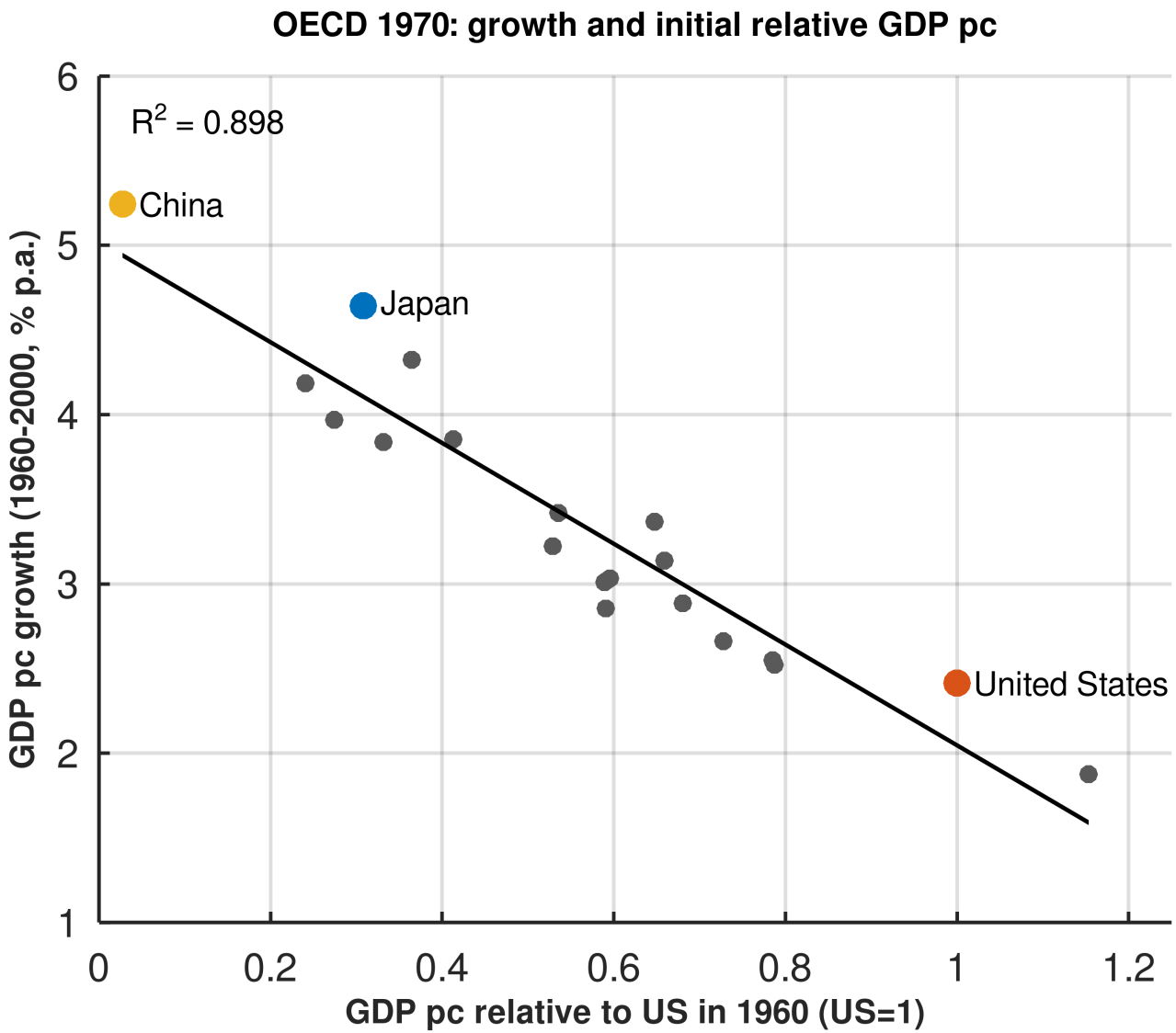

Depreciation by 2023 income group
Capital depreciation
0.05
0.045
0.04
0.035
0.03
1940
1960
1980
2000
2020
2040
Calendar year
income
High
Upper-middle
Lower-middle
Low


Log GDP pc and investment, outliers removed
R^2 = 0.449
United States
Japan
China
Zimbabwe
Log GDP pc (2000)
Average investment share, 1960-2000
12
11
10
9
8
7
6
0
0.1
0.2
0.3
0.4
0.5

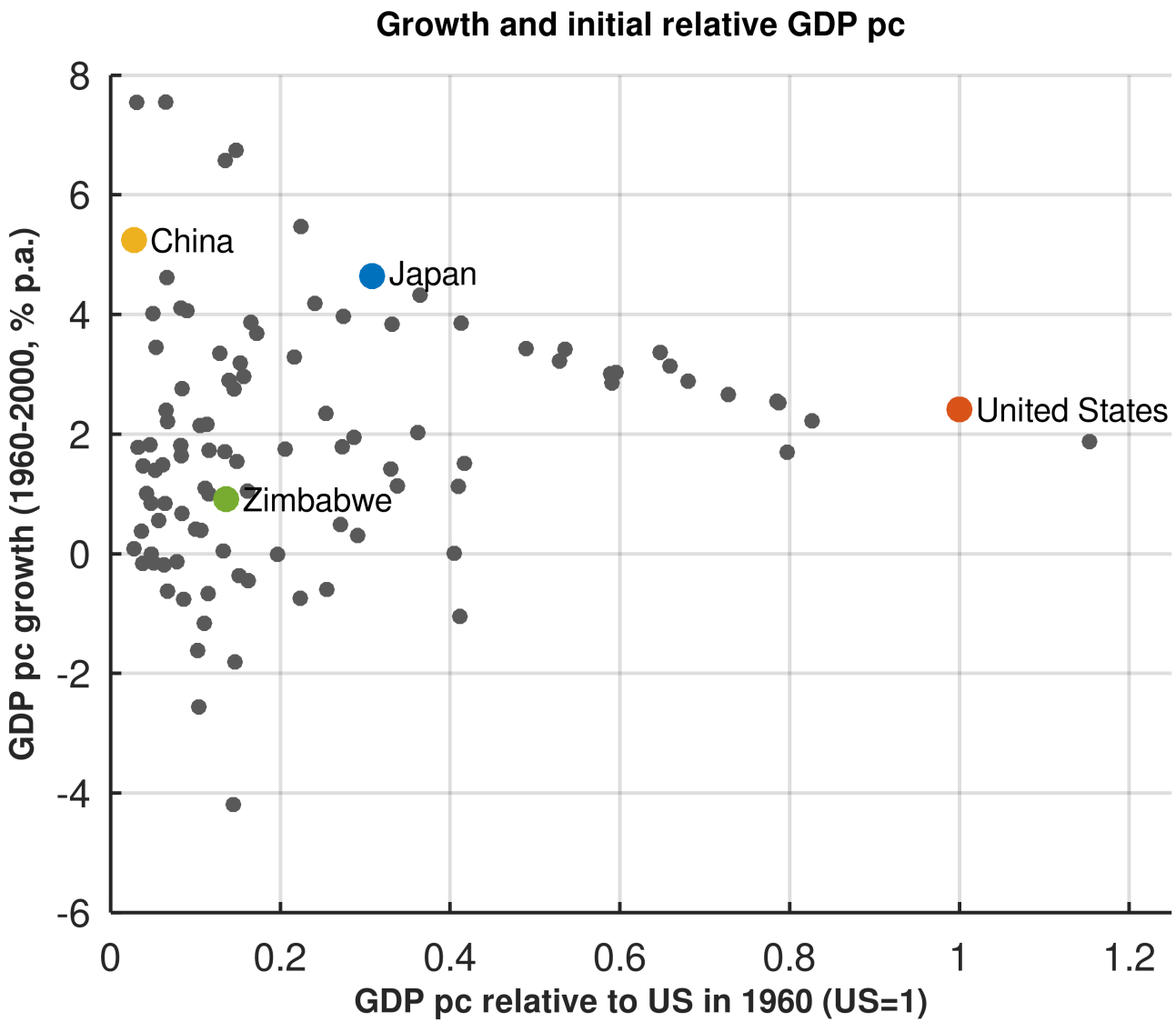

Log GDP pc and population growth, outliers removed
R^2 = 0.575
United States
Japan
China
Zimbabwe
Log GDP pc (2000)
Average population growth, 1960-2000 (percent per year)
12
11
10
9
8
7
6
0
1
2
3
4
5


Log GDP pc and depreciation, outliers removed
R^2 = 0.061
United States
Japan
China
Zimbabwe
Log GDP pc (2000)
Average capital depreciation, 1960-2000
12
11
10
9
8
7
6
0.02
0.03
0.04
0.05
0.06
0.07
0.08

Convergence hypothesis works for OECD



Convergence hypothesis does not work worldwide



The Golden Rule of Capital Accumulation

Golden rule savings rate

At the steady state of (2), $sf(k^*) = (n+\delta)k^*$, so consumption per worker is

$$c^* = (1-s)f(k^*) = f(k^*) - (n+\delta)k^*$$

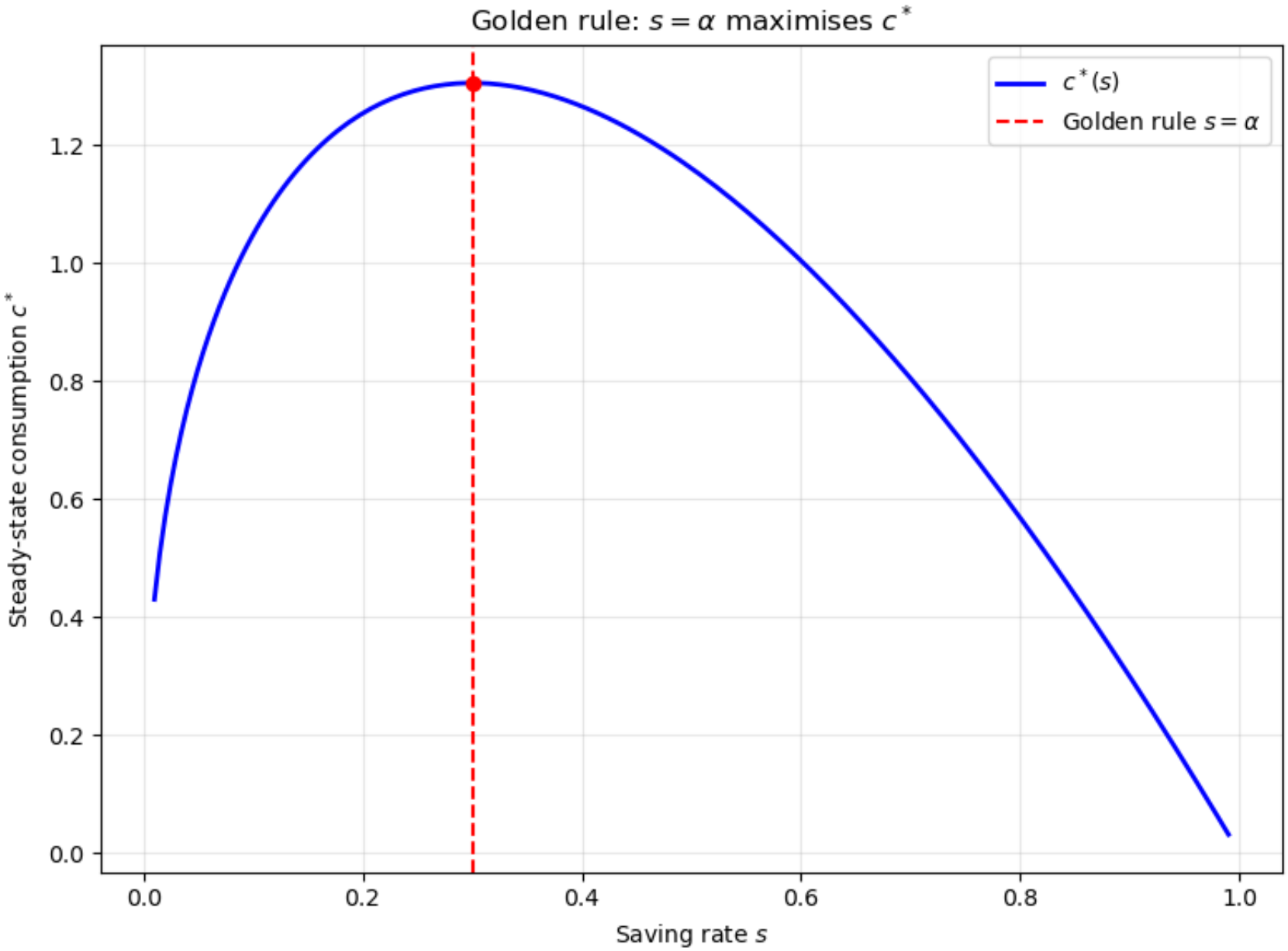
The **golden rule** k^*_{GR} maximizes c^*:

$$\frac{dc^*}{dk^*} = f'(k^*_{GR}) - (n+\delta) = 0$$

Golden rule condition

$$f'(k^*_{GR}) = n + \delta$$

For Cobb–Douglas: $\alpha k^{*\alpha-1}_{GR} = n+\delta$, so $k^*_{GR} = \left(\frac{\alpha}{n+\delta}\right)^{\frac{1}{1-\alpha}}$, which implies $s_{GR} = \alpha$.


Golden rule: $s = \alpha$ maximises c^*
$c^*(s)$
Golden rule $s = \alpha$
Steady-state consumption c^*
Saving rate s
0.0
0.2
0.4
0.6
0.8
1.0
1.2

Dynamic efficiency

- If $k^* < k^*_{GR}$ (equivalently $s < \alpha$): the economy is **dynamically efficient**.
  - Raising s increases long-run c^* but requires sacrificing current consumption.
  - There is a trade-off.
- If $k^* > k^*_{GR}$ (equivalently $s > \alpha$): the economy **over-accumulates** capital.
  - Reducing s raises consumption in *every* period—a Pareto improvement.
  - This is called **dynamic inefficiency**.

Most empirical evidence suggests real economies are dynamically efficient ($s < \alpha$).

Adding Technological Progress

Motivation

Without technological progress:

- The economy converges to a steady state with **zero growth** in per capita output.
- This contradicts the data: per capita GDP grows persistently (Kaldor SF 1).

We need to introduce **technological progress** to generate sustained per capita growth.

Labor-augmenting technology

Re-introduce technology as **labor-augmenting** (Harrod-neutral):

$$Y_t = F(K_t, A_t L_t) = K_t^{\alpha}(A_t L_t)^{1-\alpha}$$

where A_t is the level of technology growing at exogenous rate $g > 0$ per period:

$$A_{t+1} = (1+g)A_t \qquad \Rightarrow \qquad A_t = A_0(1+g)^t$$

This is equivalent to $Y_t = z_t K_t^{\alpha} L_t^{1-\alpha}$ with $z_t = A_t^{1-\alpha}$.
TFP growth is **exogenous**: the model does not explain where technology comes from.

Effective labor and new intensive variables

In the balanced growth path (BGP), all per capita variables grow at rate g. We need variables that are **constant** in BGP.
Define per **effective worker** variables:

$$\tilde{y}_t \equiv \frac{Y_t}{A_t L_t} = \frac{y_t}{A_t}, \qquad \tilde{k}_t \equiv \frac{K_t}{A_t L_t} = \frac{k_t}{A_t}$$

Then $\tilde{y}_t = \tilde{k}_t^{\alpha}$, and the fundamental equation becomes:

Solow equation with technology

$$\Delta \tilde{k}_t = \frac{s\tilde{k}_t^{\alpha} - (n + g + ng + \delta)\tilde{k}_t}{(1+n)(1+g)}$$

Since ng is tiny (e.g. $0.02 \times 0.02 = 0.0004$), $n + g + ng \approx n + g$.

Derivation of $\Delta\tilde{k}_t$

Start from capital accumulation, $K_{t+1} = sY_t + (1-\delta)K_t$, and divide by A_tL_t:

$$\frac{K_{t+1}}{A_tL_t} = s\tilde{y}_t + (1-\delta)\tilde{k}_t$$

Since $A_{t+1}L_{t+1} = (1+g)(1+n)A_tL_t$, the left side is $(1+n)(1+g)\tilde{k}_{t+1}$:

$$\tilde{k}_{t+1} = \frac{sf(\tilde{k}_t) + (1-\delta)\tilde{k}_t}{(1+n)(1+g)}$$

Subtract $\tilde{k}_t$ from both sides:

$$\begin{aligned}\Delta\tilde{k}_t &= \frac{sf(\tilde{k}_t) + (1-\delta)\tilde{k}_t - (1+n)(1+g)\tilde{k}_t}{(1+n)(1+g)} \\ &= \frac{sf(\tilde{k}_t) - (n+g+ng+\delta)\tilde{k}_t}{(1+n)(1+g)}\end{aligned}$$

Same structure as (2), with $1+n$ replaced by $(1+n)(1+g)$.

Steady state with technology

Set $\Delta\tilde{k}_t = 0$ and let $\phi \equiv n + g + ng + \delta$:

$$s\tilde{k}^{*\alpha} = \phi\,\tilde{k}^*$$

$$\tilde{k}^* = \left(\frac{s}{\phi}\right)^{\frac{1}{1-\alpha}}, \qquad \tilde{y}^* = \left(\frac{s}{\phi}\right)^{\frac{\alpha}{1-\alpha}}$$

Converting back to per worker variables:

$$k_t = A_t \cdot \tilde{k}^* = A_0(1+g)^t \left(\frac{s}{\phi}\right)^{\frac{1}{1-\alpha}}$$

$$y_t = A_t \cdot \tilde{y}^* = A_0(1+g)^t \left(\frac{s}{\phi}\right)^{\frac{\alpha}{1-\alpha}}$$

On the BGP, y_t, k_t, and c_t all grow at rate g.

Balanced growth path

Define the growth rate of x as $g_x \equiv x_{t+1}/x_t - 1$. On the BGP, $\tilde{k}_t = \tilde{k}^*$, so all per capita variables grow at rate g:

$$\frac{y_{t+1}}{y_t} = \frac{k_{t+1}}{k_t} = \frac{c_{t+1}}{c_t} = \frac{A_{t+1}}{A_t} = 1 + g$$

To verify: from $y_t = k_t^\alpha A_t^{1-\alpha}$, take the ratio of $t+1$ to t:

$$1 + g_y = (1 + g_k)^\alpha (1 + g)^{1-\alpha}$$

On the BGP, $g_k = g_y$, so:

$$(1 + g_y)^{1-\alpha} = (1 + g)^{1-\alpha} \quad \Rightarrow \quad g_y = g$$

All aggregate variables (Y_t, K_t) grow by the factor $(1+n)(1+g)$, i.e. at rate $n + g + ng \approx n + g$.

Golden rule with technology

Steady-state consumption per effective worker:

$$\tilde{c}^* = f(\tilde{k}^*) - (n + g + ng + \delta)\tilde{k}^*$$

Golden rule:

$$f'(\tilde{k}^*_{GR}) = n + g + ng + \delta$$

For Cobb–Douglas: $s_{GR} = \alpha$ (same as before).

- If $s < \alpha$: raising s increases long-run consumption per effective worker.
- If $s > \alpha$: over-accumulation—reducing s is a Pareto improvement.

Growth Accounting

The Solow residual

From $Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$, take logs and difference between t and $t+1$:

$$\Delta \ln Y_{t+1} = \Delta \ln A_{t+1} + \alpha \, \Delta \ln K_{t+1} + (1-\alpha) \, \Delta \ln L_{t+1}$$

where $\Delta \ln X_{t+1} \equiv \ln X_{t+1} - \ln X_t \approx g_X$. In growth rates:

$$g_Y = g_A + \alpha g_K + (1-\alpha) g_L$$

The **Solow residual** is the part of output growth not explained by factor accumulation:

$$g_A = g_Y - \alpha g_K - (1-\alpha) g_L$$

Also called **Total Factor Productivity (TFP)** growth. In practice, g_A accounts for a large fraction of growth—this is the part that the Solow model leaves unexplained.

Growth accounting: interpretation

- Inputs K and L are directly measurable.
- Factor shares α and $1-\alpha$ can be estimated from national accounts.
- TFP growth g_A is a **residual**—a "measure of our ignorance."

Empirical finding: TFP growth typically accounts for 50–70% of per capita output growth in developed countries. This highlights the central role of technology, which the Solow model takes as exogenous.

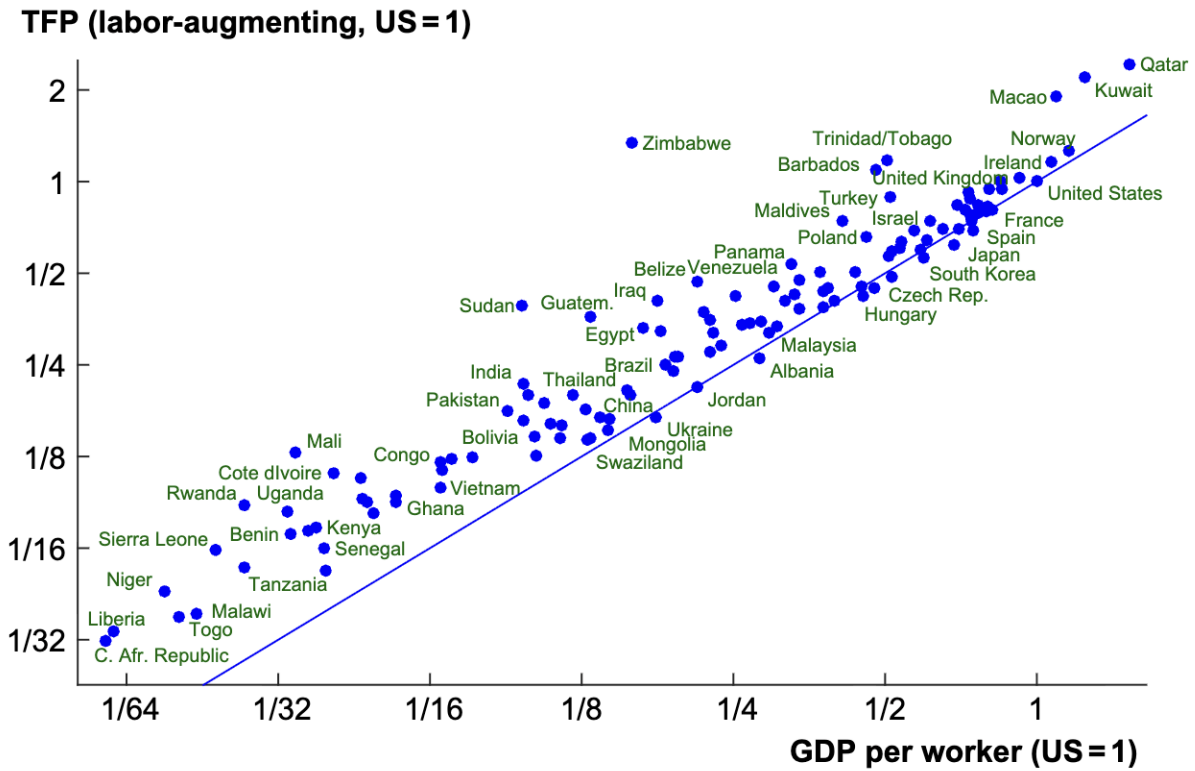


Fig. 29 Total factor productivity, 2010. Source: *Computed using the Penn World Tables 8.0 assuming a common value of* $\alpha = 1/3$.

Exercise: comparative statics in the Solow model

Consider a Solow economy with $f(k) = k^{\alpha}$, $\alpha = 1/3$, $s = 0.2$, $n = 0.02$, $\delta = 0.05$, $g = 0.02$.

1. Compute the steady-state values $\tilde{k}^*$, $\tilde{y}^*$, and $\tilde{c}^*$.
2. Compute the golden rule savings rate s_{GR} and the golden rule capital $\tilde{k}^*_{GR}$.
3. Suppose s increases permanently to $s' = 0.3$. Find the new steady state. Describe the transition path of $\tilde{k}$, $\tilde{y}$, and $\tilde{c}$ (qualitatively).
4. Compute the speed of convergence λ and the half-life $t_{1/2}$ (in years, one period = one year).
5. Is this economy dynamically efficient? Explain.

▸ Solution

Exercise: growth accounting

Suppose in a country over a 10-year period:

- Output grew at 5% per year
- Capital stock grew at 6% per year
- Labor force grew at 2% per year
- Capital share $\alpha = 0.35$

1. Compute TFP growth g_A.
2. What fraction of output growth is explained by TFP?
3. If we interpret A as labor-augmenting technology ($Y = K^{\alpha}(AL)^{1-\alpha}$), what is the growth rate of A?

APPENDIX

Solution: comparative statics exercise

$\phi = n + g + ng + \delta = 0.02 + 0.02 + 0.0004 + 0.05 = 0.0904$;
$(1+n)(1+g) = 1.0404$.

1. $\tilde{k}^* = \left(\frac{0.2}{0.0904}\right)^{3/2} = 2.212^{1.5} \approx 3.29$, $\tilde{y}^* = 3.29^{1/3} \approx 1.49$,
$\tilde{c}^* = 0.8 \times 1.49 \approx 1.19$.

2. $s_{GR} = \alpha = 1/3 \approx 0.333$. $\tilde{k}^*_{GR} = \left(\frac{1/3}{0.0904}\right)^{3/2} = 3.687^{1.5} \approx 7.08$.

3. New: $\tilde{k}^{*\prime} = \left(\frac{0.3}{0.0904}\right)^{3/2} = 3.319^{1.5} \approx 6.05$. During transition, $\tilde{k}_t$ rises gradually from 3.29 to 6.05; $\tilde{y}_t$ rises; $\tilde{c}_t$ initially *drops* (higher s, lower $1-s$) then rises as $\tilde{y}_t$ increases.

4. $\lambda = \frac{(1-1/3)(0.0904)}{1.0404} \approx 0.058$. $t_{1/2} = \ln 2 / [-\ln(1-0.058)] \approx 11.6$ years.

5. $s = 0.2 < \alpha = 1/3 = s_{GR}$, so the economy is dynamically efficient (below the golden rule).

Further Readings

- Solow, R. (1956). A contribution to the theory of economic growth. *Quarterly Journal of Economics*, 70(1), 65–94.
- Acemoglu, D. (2009). *Introduction to Modern Economic Growth*. Princeton University Press. Chapters 1–4.
- Heijdra, B. (2016). *Foundations of Modern Macroeconomics*. Chapter 12.
- Barro, R. and X. Sala-i-Martin (2004). *Economic Growth*. MIT Press. Chapters 1–2.
- Mankiw, N.G., D. Romer, and D. Weil (1992). A contribution to the empirics of economic growth. *Quarterly Journal of Economics*, 107(2), 407–437.

Solow Model with Human Capital
(Mankiw–Romer–Weil, 1992)

Motivation: the convergence speed puzzle

Recall that the gap $\tilde{k}_t - \tilde{k}^*$ shrinks by a fraction λ each period:

$$\lambda = \frac{(1-\alpha)(n+g+ng+\delta)}{(1+n)(1+g)} \approx (1-\alpha)(n+g+\delta)$$

$$t_{1/2} = \frac{\ln 2}{-\ln(1-\lambda)}$$

With $\alpha = 1/3$, $n = 0.01$, $g = 0.02$, $\delta = 0.05$: $\lambda \approx 0.05$ (half-life ≈ 13 years).
Empirical evidence: $\lambda \approx 0.02$ (half-life ≈ 35 years). The model predicts **too-fast convergence**.
Diagnosis: to slow convergence, we need a higher effective capital share α.
Solution (Mankiw, Romer, and Weil 1992, *QJE*): introduce **human capital** as a second accumulable factor. This raises the effective capital share and slows convergence.

Production with human capital

Production in period t:

$$Y_t = K_t^{\alpha_K} H_t^{\alpha_H} (A_t L_t)^{1-\alpha_K-\alpha_H}$$

with $0 < \alpha_K + \alpha_H < 1$. Here H_t is aggregate human capital, and

$$L_{t+1} = (1+n)L_t, \qquad A_{t+1} = (1+g)A_t$$

In per effective worker terms ($k_t = K_t/(A_t L_t)$, $h_t = H_t/(A_t L_t)$):

$$y_t = k_t^{\alpha_K} h_t^{\alpha_H}$$

Accumulation: households save fractions s_K and s_H of income for physical and human capital respectively:

$$K_{t+1} = s_K Y_t + (1-\delta_K)K_t$$
$$H_{t+1} = s_H Y_t + (1-\delta_H)H_t$$

Laws of motion per effective worker

Divide the accumulation equations by $A_t L_t$ and use $A_{t+1} L_{t+1} = (1+n)(1+g) A_t L_t$:

$$(1+n)(1+g)\, k_{t+1} = s_K y_t + (1-\delta_K)\, k_t$$
$$(1+n)(1+g)\, h_{t+1} = s_H y_t + (1-\delta_H)\, h_t$$

Subtract $(1+n)(1+g)k_t$ and $(1+n)(1+g)h_t$ from each side:

$$\Delta k_t = \frac{s_K k_t^{\alpha_K} h_t^{\alpha_H} - (n+g+ng+\delta_K)\, k_t}{(1+n)(1+g)}$$
$$\Delta h_t = \frac{s_H k_t^{\alpha_K} h_t^{\alpha_H} - (n+g+ng+\delta_H)\, h_t}{(1+n)(1+g)}$$

We now have a system of **two** difference equations. Same structure as (2), with n replaced by $n+g+ng$ ($\approx n+g$).

Steady state

At steady state $\Delta k_t = 0$ and $\Delta h_t = 0$. Assume $\delta_K = \delta_H = \delta$ and let $\phi \equiv n + g + ng + \delta$:

$$s_K (k^*)^{\alpha_K} (h^*)^{\alpha_H} = \phi\, k^*$$
$$s_H (k^*)^{\alpha_K} (h^*)^{\alpha_H} = \phi\, h^*$$

Dividing the two equations gives $h^*/k^* = s_H/s_K$. Substituting back:

$$k^* = \left[\left(\frac{s_K}{\phi} \right)^{1-\alpha_H} \left(\frac{s_H}{\phi} \right)^{\alpha_H} \right]^{\frac{1}{1-\alpha_K-\alpha_H}}$$

$$h^* = \left[\left(\frac{s_K}{\phi} \right)^{\alpha_K} \left(\frac{s_H}{\phi} \right)^{1-\alpha_K} \right]^{\frac{1}{1-\alpha_K-\alpha_H}}$$

Implications for convergence

Linearizing the system (in logs) around (k^*, h^*) gives a Jacobian with eigenvalues

$$\mu_1 = \frac{1-\delta}{(1+n)(1+g)}, \qquad \mu_2 = \frac{1-\delta+(\alpha_K+\alpha_H)\phi}{(1+n)(1+g)}$$

Both lie in $(0,1)$. The slow one, μ_2, governs convergence: the gap shrinks by

$$\lambda = 1-\mu_2 = \frac{(1-\alpha_K-\alpha_H)\,\phi}{(1+n)(1+g)} \approx (1-\alpha_K-\alpha_H)(n+g+\delta)$$

The effective capital share is now $\alpha_K+\alpha_H$, not just α_K. With $\alpha_K = \alpha_H = 1/3$ and the same n, g, δ:

$$\lambda = \frac{(1/3)(0.0802)}{1.0302} \approx 0.026 \qquad \Rightarrow \qquad t_{1/2} = \frac{\ln 2}{-\ln(1-\lambda)} \approx 26 \text{ years}$$

Much closer to the empirical $\lambda \approx 0.02$. Steady-state income now depends on s_K, s_H, and n.

Phase diagram with human capital

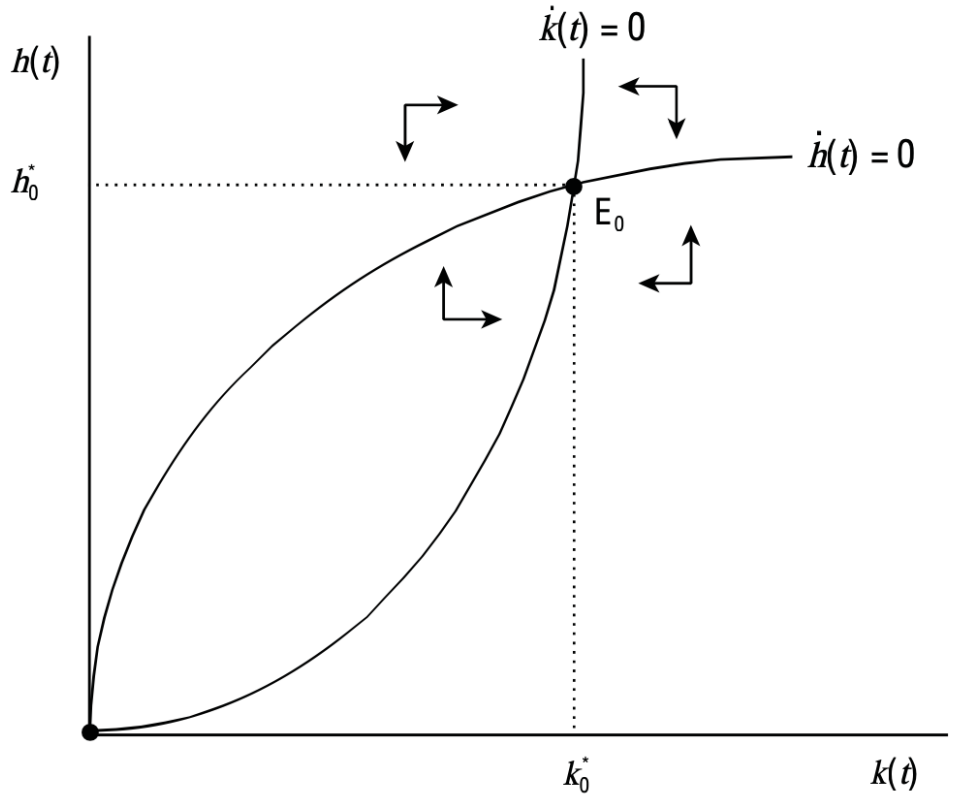


Figure 13.6: Augmented Solow-Swan model

Loci: $\Delta k_t = 0$ and $\Delta h_t = 0$. Both Jacobian eigenvalues lie in $(0, 1) \Rightarrow$ **stable node**: the system converges to (k^*, h^*) from any initial condition.